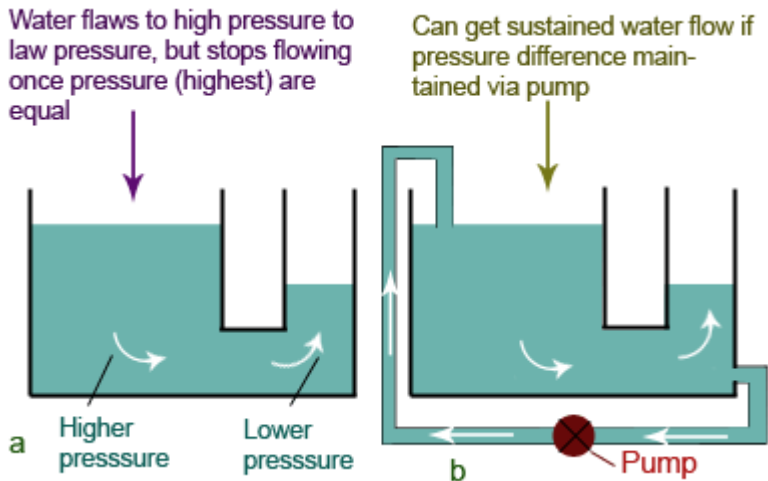


Current Electricity

- that is the **flow of electrical charge** (i.e. charge particles, usually electrons) between regions of different electrical potential
- When there is no potential, there is no charge flow (i.e. no current)
- In order to sustained current flow, need some kind of pump (i.e. battery)

Similar to fluid



What actually is flowing ???

Depends on the circuit:

In circuits of metal wires, its electrons (called conduction electrons), free to move (whereas positive ions are fixed in a lattice).

In car batteries, it's positive ions in a fluid (sulfuric acid and distilled water mix)

Current (I)

If a charge dq is flowing across a conductor in time, dt , then the current flowing through

the conductor, $I = \frac{dq}{dt}$ (units- A Ampere / A-Cs⁻¹.)

Direct current

The direction of the current does not change.



Question

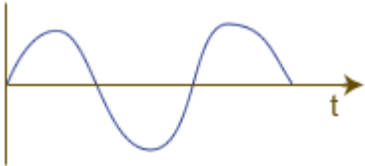
In a circuit carrying a 1-ampere current, how many electrons pass a given point in a second ?

Direct Current (dc): electrons flow in one direction only.

Eg. Battery, since terminals always have the same sign. Electrons flow from – terminal to +, and continue through circuit

Alternating current

The direction of the current changes periodically.

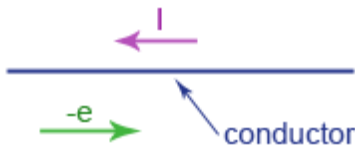


Alternating Current (ac): electrons go first in one direction, then in the opposite, then back again etc...alternating...

Eg. Household current

Conversion for current

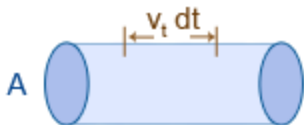
It's positive in the direction opposite to the electron flow.



Relationship between I and the properties of charge carriers

Let

- Average velocity of charge carriers = v_d (drift velocity)
- The no. of charge carriers per unit volume = n
- Charge of each carrier = q
- Total charge passing through a cross sectional area, A in time, $dt = dQ$



$$dQ = qn v_d dt A$$

$$I = \frac{dQ}{dt} = qn A v_d$$

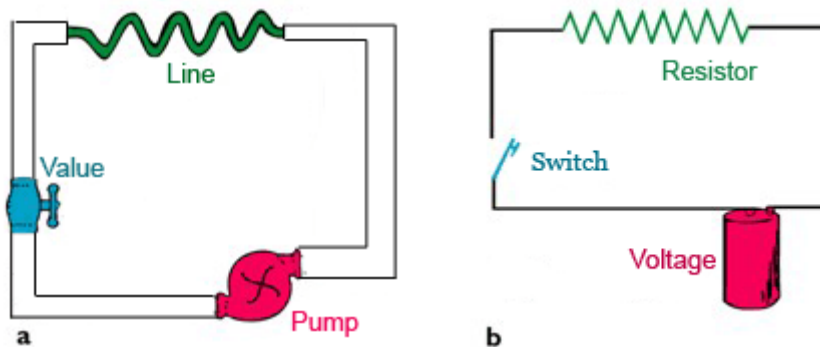
Current density,

$$J = \frac{I}{A} = qnv_d$$

$$J = qnv_d \quad (\text{units } Am^{-2})$$

Electrical Resistance:

How much current there is depends on what's in the circuit as well as the voltage. i.e on the **electrical resistance**



As shown above, electrical resistance is similar to fluids: eg. a short and wide pipe has less resistance than a long and skinny one. Similarly with wires

Free electron theory and derivation of Ohm's law

- In metals, atoms are closely packed. The outer electrons of atoms overlap. Therefore the electrons in outer orbits (valence electrons) are free to move. These free electrons are contributing to the electric current in metals.
- There are a large no. of free electrons in metals. (Eg; Cu; 8×10^{22} free electrons)
- Metals are composed of fixed charges (+ve) and free electrons moving around freely.
- While the electrons move they collide with ions and lose energy.
- The motion is random. net displacement is zero. ---- **NO CURRENT!**

But when an electric field (E) is applied electrons will experience a force (eE) and accelerate in the direction opposite to E . In this process they collide and lose energy and again accelerate.

Therefore there is an average velocity in the direction opposite to E . This velocity is called drift velocity v_d

Acceleration of electron $a = \frac{eE}{m_e} \left(\frac{\text{force}}{\text{mass}} \right) \Rightarrow m_e = \text{mass of electron}$

Let the distance between two consecutive collisions be λ (mean free path) and mean velocity of random motion be u .

Therefore time taken between two collisions, $t = \frac{\lambda}{u}$

Final velocity of electron, $v_f = \frac{\lambda}{u} \frac{eE}{m_e}$

But

$$v_d = \frac{0 + v_f}{2} = \frac{1}{2} \frac{\lambda eE}{m_e u}$$

$$v_d \propto E$$

Current density

$$J = nev_d$$

$$J = \sigma E \quad \text{where } \sigma = \frac{ne^2 \lambda}{2m_e u}$$

Where σ - conductivity

$$J = \frac{I}{A} = \frac{ne^2 \lambda}{2m_e u} \frac{v}{\ell} \quad \left\{ \begin{array}{l} dv = E dr \\ \therefore E = \frac{v}{\ell} \end{array} \right\} \Rightarrow \begin{array}{l} v = E \int dr \\ v = E \ell \end{array}$$

$$v = \left(\frac{2m_e u \ell}{ne^2 \lambda A} \right) I$$

$$V = RI \quad \text{where } R = \left(\frac{2m_e u \ell}{ne^2 \lambda A} \right)$$

$$\text{Resistance, } R = \rho \frac{\ell}{A} \quad \left\{ \rho = \frac{2m_e u}{ne^2 \lambda} \right\} \quad \rho = \text{resistivity}$$

$$\Rightarrow R = \frac{1}{\sigma} \frac{\ell}{A}$$



$$\text{Kinetic energy of electron} = \frac{1}{2} m u^2$$

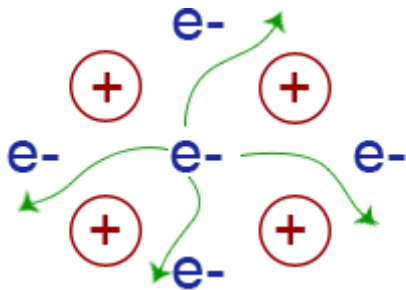
$$\text{For electrons} \quad u = \sqrt{\frac{3}{2} k T} \leftarrow \text{absolute temperature}$$

k-Boltzmaan Constant

$$\text{Where } T \uparrow \Rightarrow u \uparrow \Rightarrow R \uparrow \left(\because R = \frac{\ell}{A} \frac{2 m_e u}{n e^2 \lambda} \right)$$

$\therefore$ for metals, **increase in temperature $\Rightarrow$ Increase in resistance**

Physical interpretation



- As the temperature increases vibration of atoms and speed of electrons increases.
- $\Rightarrow$ Number of collision increases.
- $\Rightarrow$ Resistance increases (in metals)

$$R_t = R_o (1 + \alpha t + \beta t^2 + \gamma t^3 + \dots)$$

R_t - resistance at temperature t

R_o - resistance at temperature 0

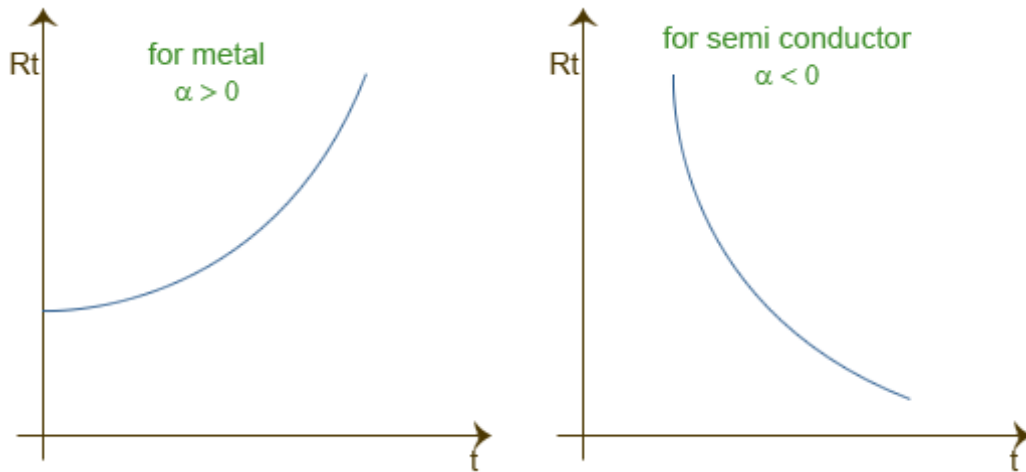
α, β, γ - constants

α, γ are very small

$$\Rightarrow R_t = R_o (1 + \alpha t)$$

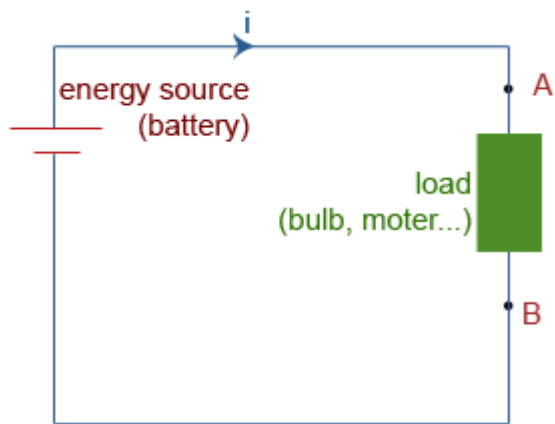
$$\alpha = \frac{R_t - R_o}{R_o t}$$

α - temperature coefficient of resistance



In semi conductors the resistance decreases with increasing temperature. Therefore thermal vibration of atoms can impart energy ($\approx KT$) to electrons breaking free from covalent bonds $\Rightarrow$ these electrons will migrate through the crystal $\Rightarrow$ current increases (resistance decreases).

Current and Energy



Suppose that a charge dq travels from $A \rightarrow B$ in time dt

$$\therefore \text{work done, } dw = dq V_{AB}$$

$$\text{But we know that } i = \frac{dq}{dt} \Rightarrow dq = i dt \Rightarrow dw = i dt V_{AB}$$

$\therefore$ work done in time t

$$\int dw = \int_0^t i dt V_{AB}$$

$$W = i V_{AB} t$$

$$\text{Work done (Energy)} = vit$$

$$\therefore \text{work done per unit time (power) } P = \frac{w}{t} = vi$$

$$P = vi$$

$$P = i^2 R \Rightarrow (v = iR)$$

Something to think ???

Electric Shock

Electric shock is caused by large currents through the body.

Since current = voltage/resistance, this also means large voltages cause shock.

But it depends on body's resistance:

If soaked with salt water, resistance can be as low as $\sim 100 \text{ } \Omega$

If skin very dry, resistance can be up to $\sim 500\,000 \text{ } \Omega$

Question

Why do we feel a tingle, when touching electrodes of a 12-V battery with moist skin, completing the circuit from one hand to the other, but we don't feel anything when our fingers are dry?

Resistors

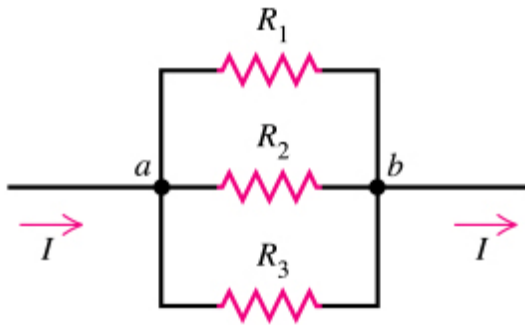
(01) Series

$$R = R_1 + R_2 + R_3$$

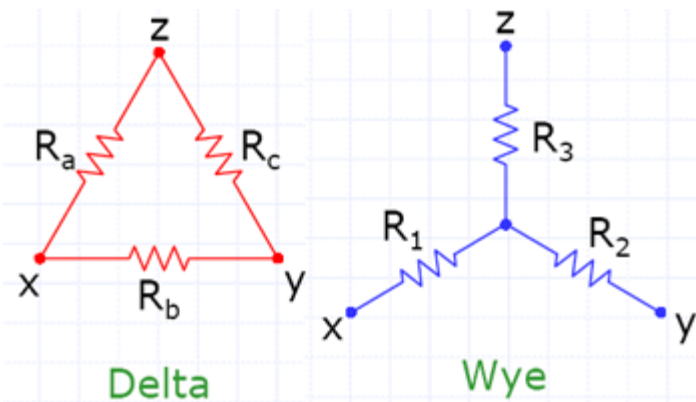


(02) Parallel

$$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$$



(3) - $\triangle$ transformation is used to establish of networks with 3 terminals

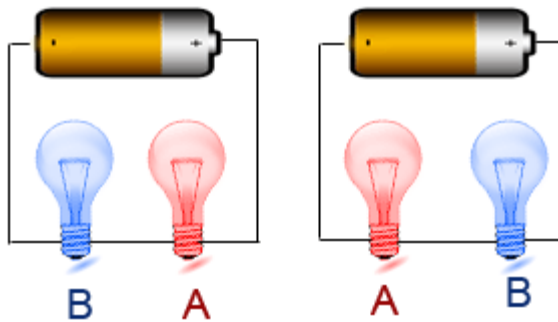


$$R_1 = \frac{R_a R_b}{R_a + R_b + R_c}$$

$$R_2 = \frac{R_b R_c}{R_a + R_b + R_c}$$

$$R_3 = \frac{R_c R_a}{R_a + R_b + R_c}$$

Question



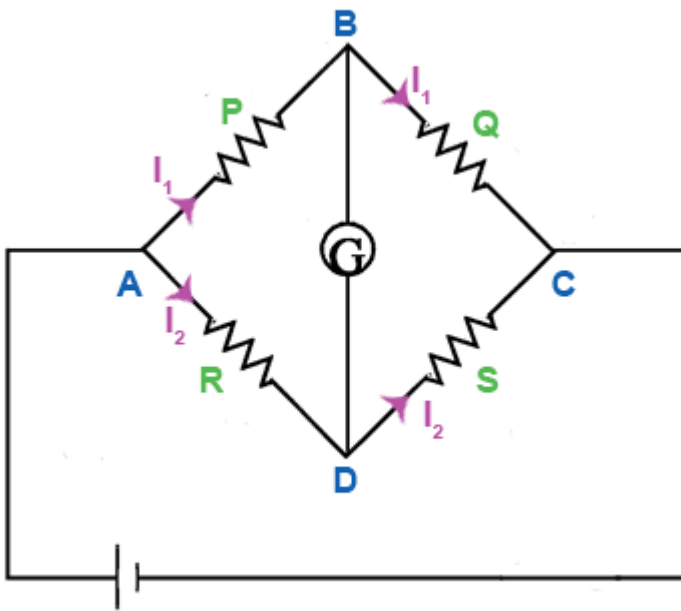
When series circuit shown to the right is connected, Bulb A is brighter than bulb B. if the positions of the bulb were reversed

- 1) Bulb A would again be brighter
- 2) Bulb B would be brighter
- 3) Either of the above occur

Measurement of resistance

Wheatstone Bridge (to measure an unknown resistance say S)

Principle



Adjust P to balance the bridge (so that $I_G = 0$)

Then

$$V_B = V_D$$

$$V_B - V_A = V_D - V_A$$

$$PI_1 = RI_2 \rightarrow (1)$$

$$\text{Similarly } V_B - V_C = V_D - V_C$$

$$QI_1 = SI_2 \rightarrow (2)$$

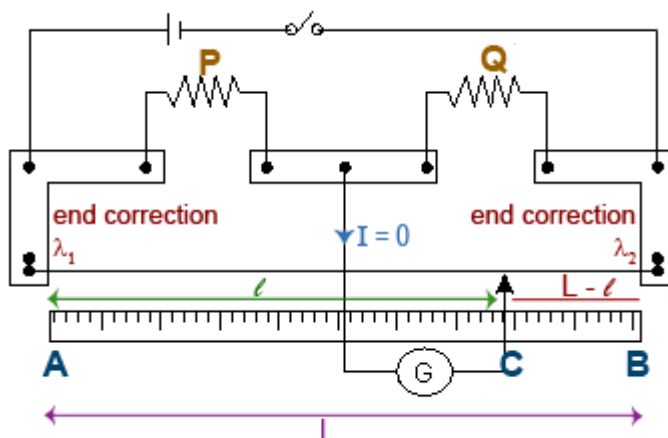
$$(1)/(2) \quad \frac{P}{Q} = \frac{R}{S}$$

$$S = \frac{QR}{P}$$

Applications of Wheatstone bridge

- (1) Meter bridge
- (2) Post-office box (to measure resistance)
- (3) Carey foster bridge

Meter bridge



$$\frac{P}{Q} = \frac{R_{AC}}{R_{CB}} \quad R_{AC} = \rho \frac{\ell}{A} \Rightarrow R_{BC} = \rho \frac{(L - \ell)}{A}$$

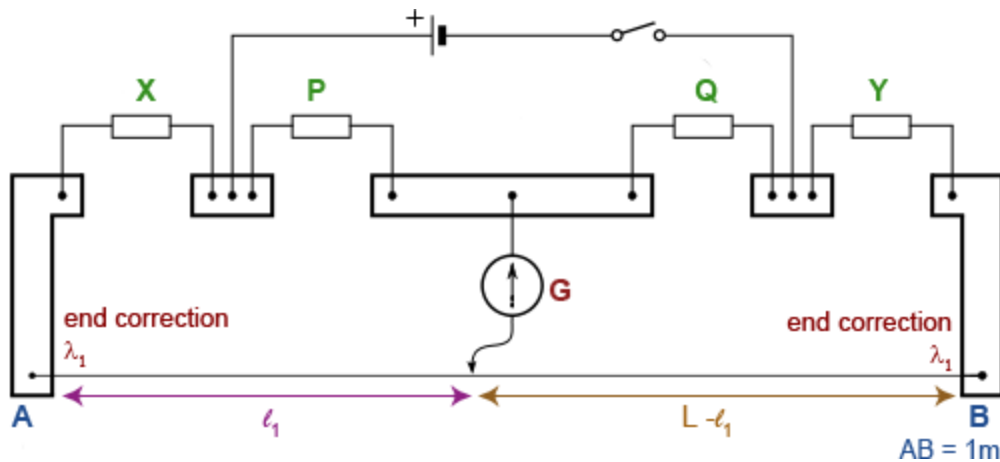
$$\frac{P}{Q} = \frac{\ell}{L - \ell}$$

When “end corrections” are incorporated

$$\frac{P}{Q} = \frac{\ell + \lambda_1}{(L - \ell) + \lambda_2} \Rightarrow \lambda_1 \text{ and } \lambda_2 \text{ should be found experimentally}$$

Carey Foster Bridge

This bridge does not need the calculation of end corrections.



When the bridge is balanced

$$\frac{P}{Q} = \frac{X + S(\ell_1 + \lambda_1)}{Y + S(L - \ell_1 + \lambda_2)} \quad S - \text{Resistance per unit length of the wire}$$

Now interchange X and Y

When the bridge is balanced let the length be ℓ_2

Then,

$$\frac{P}{Q} = \frac{Y + S(\ell_2 + \lambda_1)}{X + S(L - \ell_2 + \lambda_2)} \Rightarrow (2)$$

$$\frac{Q}{P+Q} = \frac{X + S(\ell_1 + \lambda_1)}{X + Y + S(L + \lambda_1 + \lambda_2)} \Rightarrow (A)$$

$$\frac{P}{P+Q} = \frac{Y + S(\ell_2 + \lambda_1)}{X + Y + S(L - \lambda_1 + \lambda_2)} \Rightarrow (B)$$

$$(A) \text{ and } (B) \Rightarrow X + S(\ell_1 + \lambda_1) = Y + S(\ell_2 + \lambda_1)$$

$$\Rightarrow X = Y + S(\ell_2 - \ell_1)$$

X – unknown resistor

Y – known resistor (eg; resistance box)

S – can be calculated using known resistors for x and y

Carey Foster bridge is especially used to measure the temperature coefficient of resistance of materials accurately.

Also it can be used to find the resistance per unit length of a given wire (s).

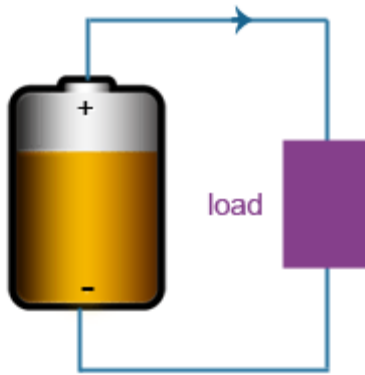
Electromotive force (emf)

Electromotive force is the difference in electric potential (voltage) between terminals of a source of electricity (eg. A battery) from which **no current** is being drawn.

Consider an ordinary dry cell as a source of electrical energy.



Chemical reactions occur in the dry cell develop a **potential difference** between the positive and negative terminals.

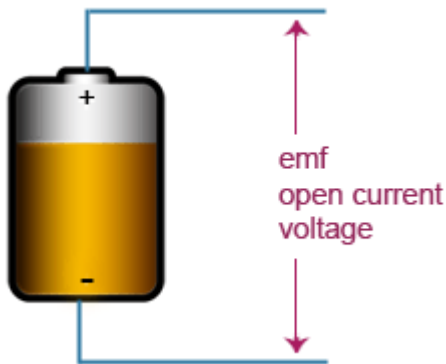


If a current path is provided, current can flow throughout the air unit (closed air unit).

When the current flows through the battery (closed air unit) it has to overcome the **internal resistance** of the battery → some of the energy of the battery is used.

The emf of a battery is usually defined as the open air unit (no current path) potential across the terminals.

∴ closed air unit voltage across a battery is always less than the emf (open air unit voltage) of the battery.



emf is measured in volts (V)

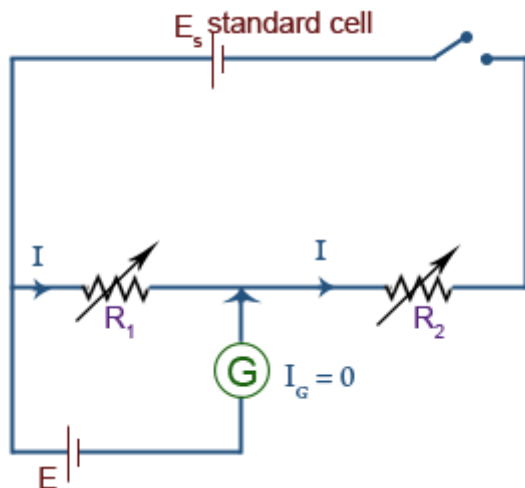
Measurement of emf

What is a standard cell?

It is a primary cell whose voltage is accurately known and remains **sufficient constant** for instrument calibration purposes.

Eg. Weston Standard cell has a voltage of
(1.018636 V at 20°C)

Rayleigh Potentiometer



E_s – Standard cell (with known emf)

R_1, R_2 – variable resistors

G – Galvanometer

E – Battery of which the emf to be measured.

- Adjust R_1, R_2 So that the current through the Galvanometer = 0 (i.e. $I_G = 0$)
- At this point

$$E = IR_1 \quad (1)$$

$$E_s = I(R_1 + R_2) \quad (2)$$

$$(1) \text{ \& } (2) \rightarrow E = \frac{E_s R_1}{(R_1 + R_2)} \quad (A)$$

- If you want to compare this E with another unknown emf (E' - say), replace E with E' and find new R_1' & R_2' at there, similarly you may find

$$E' = \frac{E_s R_1'}{(R_1' + R_2')} \quad (B)$$

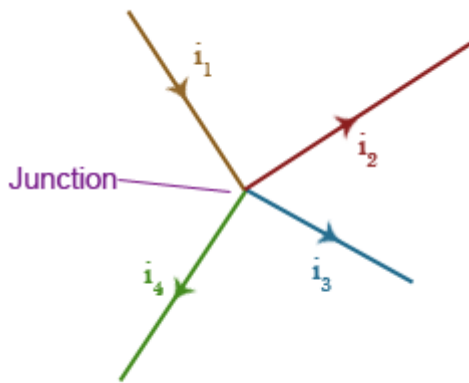
(A) & (B) $\rightarrow$

$$\frac{E}{E'} = \frac{R_1 (R_1' + R_2')}{(R_1 + R_2) R_2'} \quad (C)$$

- In the two experiments it's possible to keep $(R_1 + R_2) = (R_1' + R_2')$

$$(C) \rightarrow \frac{E}{E'} = \frac{R_1}{R_2'}$$

Kirchhoff's Laws



Sign conversion

Flowing in is positive

Flowing out is negative

(O1) Junction rule

The net current into any junction equals the current out of the junction

i. e. $i_1 - i_2 - i_3 - i_4 = 0$

$$\sum i = 0$$

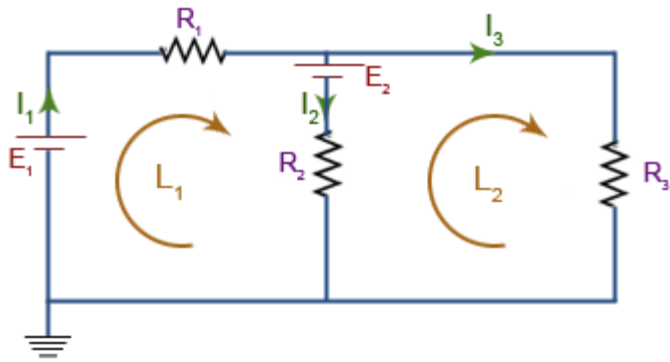
(2) loop rule

The algebraic sum of the *emf* ($\sum E$) equals the algebraic sum of the potential drop due to resistance $\sum IR$ around the loop

i.e. $\sum E = \sum IR$ - (unless otherwise stated neglect the internal resistance of the battery)



eg :



$$I_1 = I_2 + I_3 \quad (1)$$

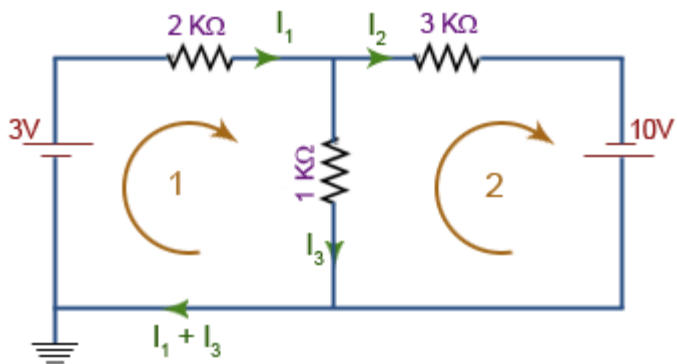
$$L_1 \quad E_1 - E_2 = I_1 R_1 + I_2 R_2 \quad (2)$$

$$L_2 \quad E_2 = I_3 R_3 - I_2 R_2 \quad (3)$$

→ I_1, I_2 & I_3 Can be found!

(Known E_1, E_2, R_1, R_2 & R_3)

problem



Loop rule

$$(1) \quad 3 = 2I_1 + I_3 \quad (1)$$

$$(2) \quad 10 = 3I_2 - I_3 \quad (2)$$

Junction rule

$$I_1 = I_2 + I_3 - (3)$$

$$(1) \rightarrow 3 = 2(I_2 + I_3) + I_3 - (4)$$

$$(2) \rightarrow I_2 = \frac{10 + I_3}{3} - (5)$$

Substitute in (4)

$$3 = 2 \left[\frac{(10 + I_3)}{3} + I_3 \right] + I_3$$

$$9 = 2 [10 + I_3 + 3I_3] + 3I_3$$

$$9 = 20 + 11 I_3$$

$$I_3 = -1mA$$

$$(5) \rightarrow I_2 = 3mA$$

$$(3) \rightarrow I_1 = -2mA$$

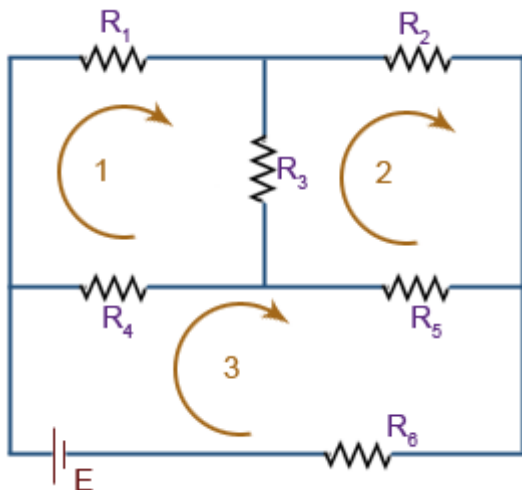
Maxwell's cyclic currents.

Equations obtained by the application of Kirchhoff's laws are sometimes quite dull in a complicated network.

→ Maxwell suggested a simple method called cyclic currents method.

In this method a certain magnitude of current is assumed to flow in each mesh - all the cyclic currents being in the same direction.

(i.e. either clockwise or anti - clockwise and then apply Kirchhoff's loop law)



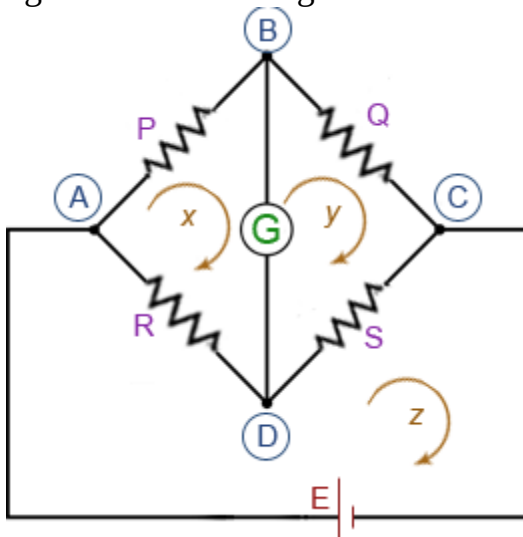
→ i_1, i_2, i_3 unknown !

by 3 equations

→ i_1, i_2, i_3 can be found ?

Applications of Maxwell's Cyclic Current

Eg. Wheatstone bridge



Resistance of $G = R_g$

ABDA

$$xp + (x - y) R_g + (x - z) R = 0 - (1)$$

BCDB

$$yQ + (y - z) S + (y - x) R_g = 0 - (2)$$

$$x(p + R_g + R) - y(R_g) - zR = 0 - (3)$$

$$-x(R_g) + y(Q + S + R_g) - zS = 0 - (4)$$

$$(3) \times S - (4) \times R$$

$$x[s(p + R + R_g) + RR_g] - y[SR_g + R(Q + S + R_g)] = 0 - (5)$$

The current in the galvanometer = $x - y$

At the balance point $(x - y) = 0 \rightarrow x = y$

$$\rightarrow s(p + R + R_g) + RR_g = R(Q + S + R_g) + SR_g$$

$$SP = RQ$$

$$P/Q = R/S$$